

Ultra-High Optical Output Power External Laser Sources for Co-Packaged Optics

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Abstract: We describe the design and characteristics of an 8-channel TOSA employing high-power SOA-integrated DFB-LDs. We also describe a 16-channel ELSFP module integrated with the two TOSAs.

1. Introduction

The rapid growth of artificial intelligence (AI) and cloud computing has dramatically increased data center traffic. To meet the demands for increasing bandwidth and enhancing energy efficiency, the adoption of Co-Packaged Optics (CPO) to data center networks has been expected. Silicon Photonics (SiPh) transceivers are placed near a large-scale integration circuit (LSI) such as a switch application specific integrated circuit (ASIC) or a graphics processing unit (GPU) on the same substrate [1]. In this packaging architecture, the heat generated by adjacent LSI raises an operating temperature of SiPh transceivers, thereby potentially affecting reliability and characteristics of integrated semiconductor lasers. To address this concern, external laser sources separated from the SiPh transceiver were proposed. The Optical Internetworking Forum (OIF) has published an implementation agreement (IA) for the external laser small form factor pluggable (ELSFP) module [2]. ELSFP modules are located at a front panel where an ambient temperature is sufficiently low, and it can be easily replaced if it is failed. The OIF ELSFP IA specifies optical power classes and channel count options. We have demonstrated a +21-dBm $\times$ 16-channel uncooled CWDM ELSFP module employing two 8-channel TOSAs [3]. There is a demand to increase fiber-coupled optical power of $>+23$ dBm to support 200-Gb/s and higher per channel SiPh transceivers. We newly designed a higher optical power semiconductor optical amplifier (SOA)-integrated distributed feedback (DFB)-laser diode (LD) to achieve fiber-coupled power of $>+25$ dBm, corresponding to the super high power (SHP) class [4].

In this paper, we describe two different SOA-integrated DFB-LDs designed to support different optical power classes. We also describe the design and characteristics of 8-channel TOSAs employing the two different SOA-integrated DFB-LDs. The cooled TOSA achieves fiber-coupled power of $>+25$ dBm per channel with the higher optical power SOA-integrated DFB-LDs. We also report a 16-channel ELSFP module employing the two TOSAs.

2. SOA-integrated DFB-LD

To increase optical power, we adopted an SOA region which is connected to a DFB region via a spot size converter. The SOA region features a wider mesa width than that of the DFB region, which contributes reducing electrical and thermal resistances to realize high optical output power. The SOA-integrated DFB-LD employs InP-based materials and a GaInAsP multiple quantum well structure which is grown by using the metal-organic chemical vapor deposition (MOCVD) method. The DFB-LD adopts a buried heterostructure (BH) configuration, in which the active region is embedded within InP cladding layers to enhance current confinement and thermal stability.

The OIF ELSFP IA specifies optical power classes where very high power (VHP), ultra-high power (UHP), and SHP are defined as $+20 \pm 1.5$ dBm, $+23 \pm 1.5$ dBm, and $+26 \pm 1.5$ dBm, respectively. First, we designed and fabricated an

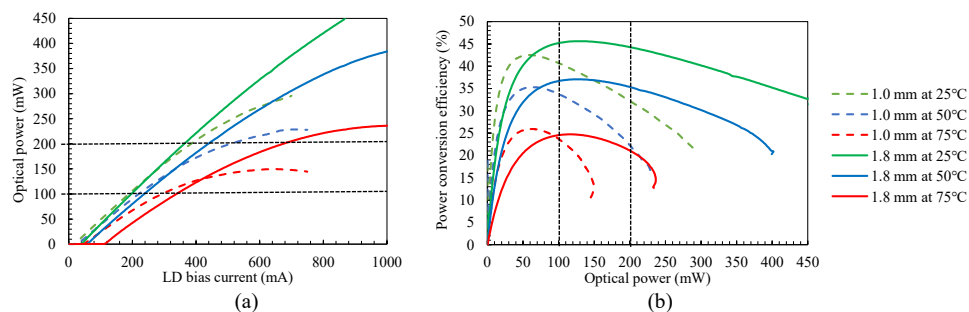


Fig. 1 IL-curves and PCEs as function of optical power for 1.0mm and 1.8mm cavity DFB-LD, (a) IL-curves. (b) PCEs.

SOA-integrated DFB-LD with a cavity length of 1.0 mm, which is optimized for the VHP ELSFP module operated in air-cooling conditions. We demonstrated a +21-dBm $\times$ 16-channel ELSFP module with the 1.0-mm long SOA-integrated DFB-LD. In order to support the VHP and SHP classes, we also designed and fabricated a 1.8-mm long SOA-integrated DFB-LD. Fig. 1 shows characteristics of optical power and power conversion efficiency (PCE) as a function of optical output power for 1.0-mm and 1.8-mm long SOA-integrated DFB-LDs, respectively. We assembled an SOA-Integrated DFB-LD to an aluminum nitride (AlN) substrate. An operating temperature is sensed at the bottom surface of the substrate. As shown in Fig. 1 (a), the 1.0-mm SOA-integrated DFB-LD realize $>+20$ dBm (>100 mW) over an operating temperature range of 25 °C to 75 °C and an LD bias current of 350 mA and higher. The 1.8-mm SOA-integrated DFB-LD reaches +26 dBm (400mW) at an operating temperature of 25 °C and an LD bias current of 750 mA. It also realizes $>+23$ dBm (>200 mW) over an operating temperature range of 25 °C to 75 °C and an LD bias current of 700 mA and higher. As shown in Fig. 1 (b), PCE characteristics are better for the 1.0-mm long SOA-integrated DFB-LD within 100 mW. The 1.8-mm long SOA-integrated DFB-LD has better PCE characteristics in the optical power range of >100 mW.

3. 8-channel TOSA employing SOA-integrated DFB-LDs

Fig. 2 shows a schematic illustration for the 8-channel TOSA structure. An 8-channel ELS module integrates an 8-channel TOSA and its control circuitries inside a housing. Hence, a mechanical size of the 8-channel TOSA is as small as 22.5 mm $\times$ 13 mm $\times$ 3.3 mm. The SOA-integrated DFB-LDs are mounted on an AlN substrate with high thermal conductivity [5]. To suppress thermal interference from adjacent LDs, a mounting pitch of SOA-integrated DFB-LDs was specified to 1.2 mm. The continuous wave (CW) laser beams are coupled to polarization-maintaining fibers (PMFs) by a parallel optical coupling system employing focusing-lens-attached optical isolators. In the optical coupling assembly process, coupling efficiencies and polarization extinction ratios (PERs) are maximized by aligning optical axes and rotating an angle of the PMF. Then, the PMF is fixed with an adhesive material. All channels have a typical coupling efficiency of $\sim 80\%$ and a PER of >20 dB.

Fig. 3 shows simulated LD temperature characteristics for 1.0-mm and 1.8-mm long SOA-integrated DFB-LDs at a case temperature of 55 °C. To ensure the reliability of SOA-integrated DFB-LDs, it is preferable that each LD temperature is <65 °C during the service. Hence, the maximum allowable LD bias current for the 1.0-mm long SOA-integrated DFB-LDs is estimated to 600 mA at a case temperature of 55 °C. Similarly, the maximum allowable LD bias current is estimated to 870 mA for the 1.8-mm long SOA-integrated DFB-LD.

Fig. 4 shows optical output power characteristics of the 8-channel TOSAs at case temperatures of 25 °C and 55 °C. Dotted and solid curves show the characteristics when employing 1.0-mm and 1.8-mm long SOA-integrated DFB-LDs, respectively. The 8-channel TOSA employing the 1.0-mm long DFB-LDs achieves optical output power of $>+20$

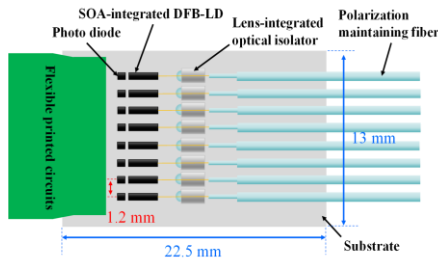


Fig. 2 Schematic illustration for the 8-channel TOSA structure.

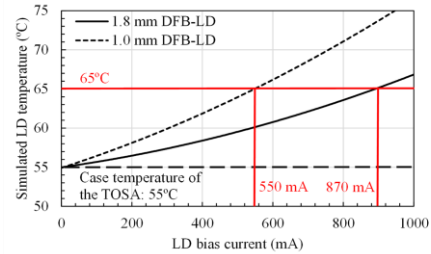


Fig. 3 Simulated LD temperature at a case temperature of 55 °C

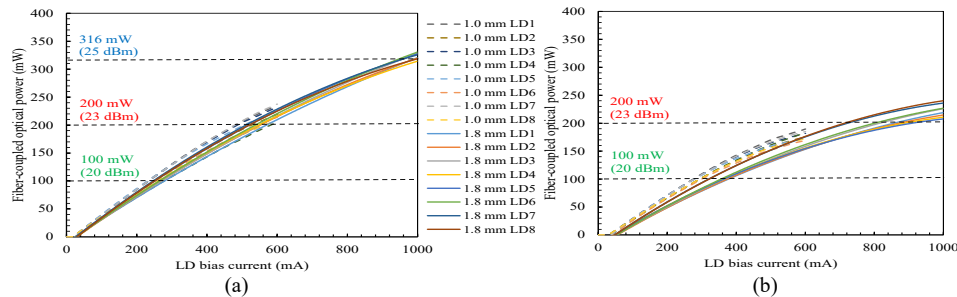


Fig. 4 Fiber-coupled optical power characteristics as a function of LD bias current for the 8-channel TOSAs employing 1.0-mm and 1.8-mm cavity DFB-LD at case temperature of 25 °C and 55 °C. (a) 25 °C. (b) 55 °C.

dBm for all channel at both case temperatures. At optical power of +20 dBm, PCEs measure 23.5 % and 21.6 % at case temperatures of 25 °C and 55 °C, respectively. The 8-channel TOSA employing 1.8-mm long SOA-integrated DFB-LDs achieves 25 dBm and higher at a case temperature of 25 °C. This indicates that this TOSA can achieve optical power of >+25 dBm in a cooled condition. The 1.8-mm SOA-integrated DFB-LD can allow an LD current exceeding 1000 mA while enduring the reliability. At a case temperature of 55 °C, the TOSA 1.8-mm long SOA-integrated DFB-LDs realize +23 dBm around an LD bias current of 870 mA for all channel. Therefore, this TOSA can be adopted for uncooled UHP and cooled SHP ELSFP modules.

4. 16-channel ELSFP module

To demonstrate the maximum channel number specified in the OIF ELSFP IA, we designed and fabricated a 16-channel ELSFP module. The ELSFP module contains two 8-channel TOSAs, a printed circuit board assembly (PCBA), and a blind mate adaptor, and a housing. Fig. 5 shows a schematic illustration of the internal structure of the 16-channel ELSFP module. The PCBA includes a power supply regulator, a microcontroller unit (MCU), a temperature sensor, and laser driver circuits which enable automatic power and current control for 16 SOA-integrated DFB-LDs. Two MT ferrules are connected to the blind mate adaptor which is positioned at the card edge. All PMFs of the TOSA are routed within the housing with a bend radius of >7.5 mm to prevent increase of fiber bending loss [5]. Fig. 6 shows schematic illustrations of an MT ferrule connected to PMFs. To ensure PER at the optical interface, an alignment line passing through the centers of stress rods and a core, as a slow axis, is precisely aligned to a target line passing through the centers of the two guide holes. We specified a rotation angle error as an angular difference between the target line and the alignment line.

Table 1 summarizes measured rotational angle errors and PER values for all channels. The ELSFP module achieved rotational angle errors of <2 degree and PERs of >20 dB for all channels.

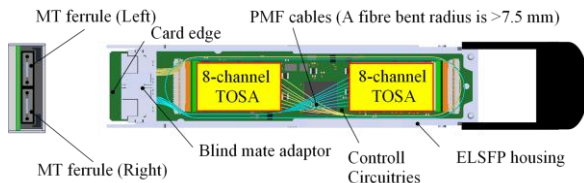


Fig. 5 Schematic illustrations of internal structure of a 16-channel ELSFP module.

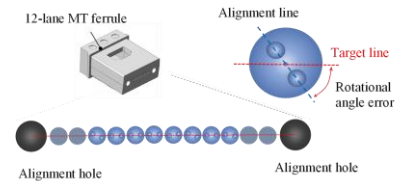


Fig. 6 Schematic illustration of the MT ferrule.

Table 1: Measured rotational angle error and PER

Position	MT ferrule (left)								MT ferrule (right)							
	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
Rotational angle error (degree)	1.2	0.5	0.1	0.4	0.1	0.6	0.4	0.9	0.5	0.4	0.9	0.9	0.6	1.3	1.4	1.6
PER (dB)	23.2	23.7	23	24.8	23.8	24.4	22.7	23.3	24.6	23.9	24.3	25.4	28.5	22.4	22.5	25.2

5. Conclusion

We described the 1.0-mm and 1.8-mm long SOA-integrated DFB-LDs designed to support different optical power classes of ELSFP modules. We also described the design and characteristics of 8-channel TOSAs employing the two different SOA-integrated DFB-LDs. The TOSA employing the 1.8-mm long SOA-integrated DFB-LDs achieves fiber-coupled power of >+25 dBm per channel in a cooled condition and >+23 dBm per channel in an uncooled condition. The TOSA employing the 1.0-mm long SOA-integrated DFB-LDs is suitable for >+20 dBm per channel in an uncooled condition. We also described a 16-channel ELSFP module employing the two TOSAs. To supply linearly polarized laser beams to SiPh transceivers, all PMFs are precisely aligned and fixed to overlap the alignment line and the target line on the facet of an MT ferrule. The ELSFP module realized rotational angle errors of <2 degree and PERs of >20 dB for all channels.

6. References

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